Rectifier



Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

1 labs

Covers these lab items: Rectifier

What it is used for

A rectifier converts alternating current into direct current for suitable electronic circuits or power applications. Identify the rectifier type and verify its electrical ratings before connecting it to a circuit.

How to use

1. Inspect the rectifier, terminals, heat sink, and package markings for damage.
2. Read the datasheet to confirm input limits, output ratings, polarity, and pin configuration.
3. Disconnect power before installing or wiring the rectifier.
4. Connect AC input and DC output terminals according to the circuit diagram.
5. Check polarity, insulation, and load suitability before energizing the circuit.
6. Apply the specified input and measure the DC output using a suitable meter.
7. Switch off and isolate the supply before inspecting or changing connections.

⚠ Safety

- Never exceed the rectifier's voltage, current, or temperature ratings.
- Avoid contact with exposed live terminals and charged capacitors.
- Use suitable fuses, insulation, and heat dissipation where required.
- Only trained personnel should work on hazardous mains-voltage circuits.

Care & storage

- Allow components to cool, inspect terminals, and store spare rectifiers in dry antistatic packaging.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.